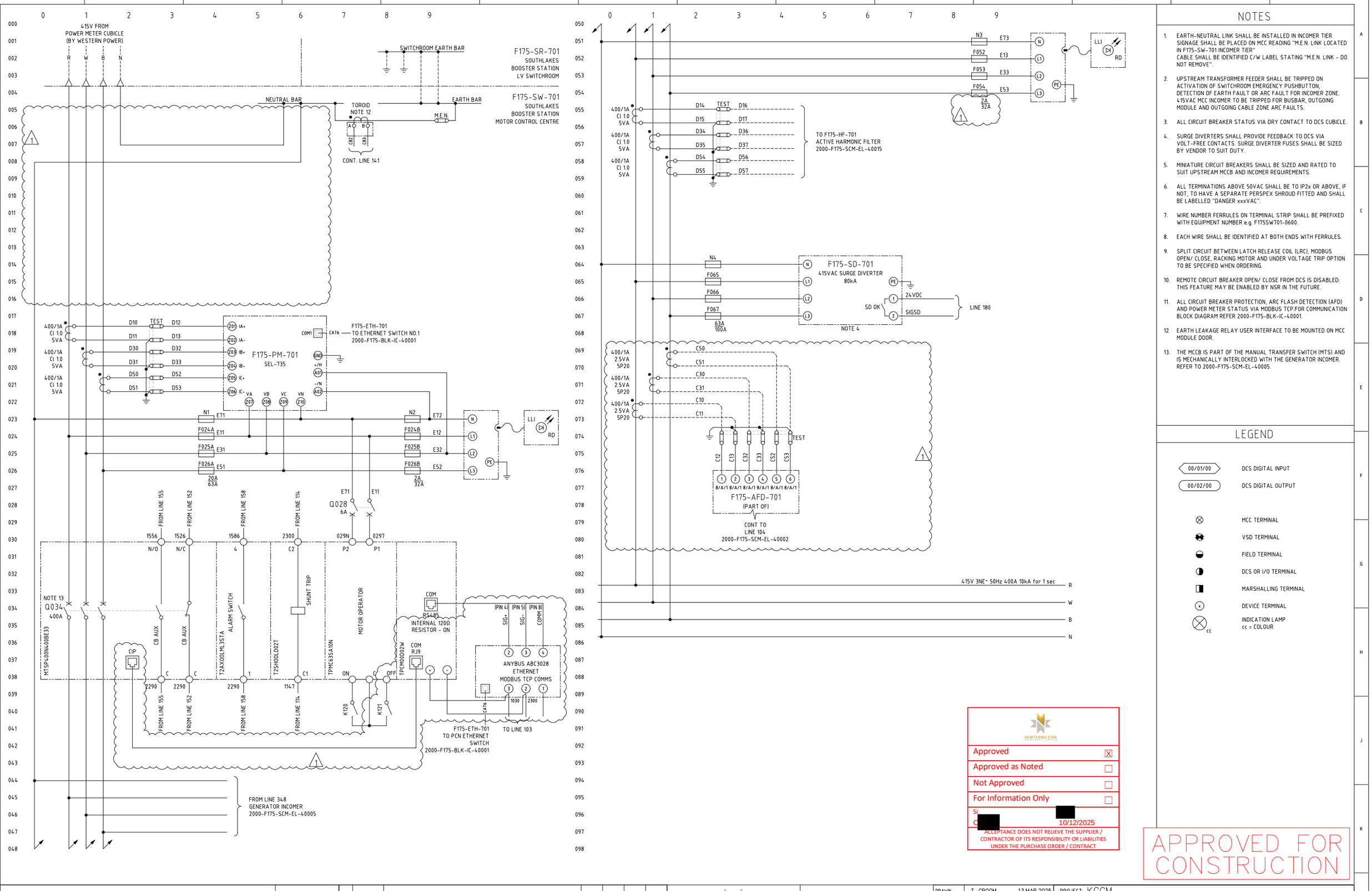



## NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC READING "M.E.N. LINK LOCATED IN F175-SW-701 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT FOR INCOMER ZONE. 415VAC MCC INCOMER TO BE TRIPPED FOR BUSBAR, OUTGOING MODULE AND OUTGOING CABLE ZONE ARC FAULTS.
3. ALL CIRCUIT BREAKER STATUS VIA DRY CONTACT TO DCS CUBICLE.
4. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
5. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT UPSTREAM MCCB AND INCOMER REQUIREMENTS.
6. ALL TERMINATIONS ABOVE 50VAC SHALL BE TO IP2x OR ABOVE, IF NOT, TO HAVE A SEPARATE PERSPEX SHROUD FITTED AND SHALL BE LABELLED "DANGER xxxVAC".
7. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F175SW701-0600.
8. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
9. SPLIT CIRCUIT BETWEEN LATCH RELEASE COIL (LRC), MODBUS OPEN/ CLOSE, RACKING MOTOR AND UNDER VOLTAGE TRIP OPTION TO BE SPECIFIED WHEN ORDERING.
10. REMOTE CIRCUIT BREAKER OPEN/ CLOSE FROM DCS IS DISABLED: THIS FEATURE MAY BE ENABLED BY NSR IN THE FUTURE.
11. ALL CIRCUIT BREAKER PROTECTION, ARC FLASH DETECTION (AFD) AND POWER METER STATUS VIA MODBUS TCP.FOR COMMUNICATION BLOCK DIAGRAM REFER 2000-F175-BLK-IC-40001.
12. EARTH LEAKAGE RELAY USER INTERFACE TO BE MOUNTED ON MCC MODULE DOOR.
13. THE MCCB IS PART OF THE MANUAL TRANSFER SWITCH (MTS) AND IS MECHANICALLY INTERLOCKED WITH THE GENERATOR INCOMER. REFER TO 2000-F175-SCM-EL-40005.

## LEGEND

- 00/01/00 — DCS DIGITAL INPUT
- 00/02/00 — DCS DIGITAL OUTPUT
- MCC TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL
- INDICATION LAMP (cc = COLOUR)

| | Approved | Approved as Noted | Not Approved | For Information Only |
|---|---|---|---|---|
| | ☒ | ☐ | ☐ | ☐ |

Si [redacted] 10/12/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

APPROVED FOR CONSTRUCTION

| REFERENCES | DWG No | REV | DATE | REVISION | DRN | CHK | TECH APP | PROJ APP |
|---|---|---|---|---|---|---|---|---|
| F175-SW-701 LV MOTOR CONTROL CENTRE INCOMER SCHEMATIC DIAGRAM SHEET 3 | 2000-F175-SCM-EL-40005 | | | | | | | |
| F175-SW-701 LV MOTOR CONTROL CENTRE INCOMER SCHEMATIC DIAGRAM SHEET 2 | 2000-F175-SCM-EL-40002 | 1 | 04.DEC.25 | RE-ISSUED FOR CONSTRUCTION | Initial ET TG | DS MC JYW | DS KVV KVN | DS EH TH |
| F175-SW-701 LV MOTOR CONTROL CENTRE SINGLE LINE DIAGRAM | 2000-F175-SLD-EL-40001 | 0 | 21.MAY.25 | APPROVED FOR CONSTRUCTION | | | | |

NORTHERN STAR RESOURCES LIMITED

DRA PACIFIC (PTY) LIMITED

| | | |
|---|---|---|
| DRAWN | T. GROOM | 13.MAR.2025 |
| CHECKED | J.Y. WEE | 01.APR.2025 |
| DESIGNED | J.Y. WEE | 01.APR.2025 |
| TECH APP | K. VAN NAMEN | 01.APR.2025 |
| PROJ APP | T. HAMDORF | 21.MAY.2025 |
| CLIENT APP | | |

PROJECT: KCGM
FIM TSF PROJECT

TITLE: SOUTHLAKES BOOSTER STATION
F175-SW-701 LV MCC MAIN INCOMER
SCHEMATIC DIAGRAM SHEET 1

A1 | SCALE NTS | DRG No. 2000-F175-SCM-EL-40001 | REV 1